

FUNDAMENTAL QUANTITIES (DETAILED NOTES)

1. Definition:

Fundamental quantities are the basic physical quantities that do not depend on any other quantities.

They form the foundation of all physical measurements and are used to derive other quantities.

2. Importance:

- Provide a standard system of measurement (SI system)
- Used to express all physical laws
- Help avoid confusion in measurements worldwide

3. The Seven Fundamental Quantities:

1) Length (meter, m)

Definition: Distance between two points.

Examples: height of a person, distance between cities.

2) Mass (kilogram, kg)

Definition: Amount of matter in a body.

Note: Mass remains constant everywhere.

Examples: mass of a book, rice, or any object.

3) Time (second, s)

Definition: Duration between two events.

Examples: time taken to travel, cooking time.

4) Electric Current (ampere, A)

Definition: Flow of electric charge.

Examples: current in wires, charging devices.

5) Temperature (kelvin, K)

Definition: Measure of hotness or coldness.

Concept: Indicates kinetic energy of particles.

6) Amount of Substance (mole, mol)

Definition: Number of particles in a substance.

1 mole = 6.022×10^{23} particles.

7) Luminous Intensity (candela, cd)

Definition: Brightness of a light source.

Examples: brightness of bulbs and LEDs.

4. Key Characteristics:

- Independent (not derived from other quantities)
- Universal (same everywhere)
- Standardized (SI units)
- Measurable using instruments

5. Relation to Derived Quantities:

Derived quantities are formed using fundamental quantities.

Examples:

Velocity = Length / Time

Force = Mass $\times$ Acceleration

6. Conclusion:

Fundamental quantities are the building blocks of physics. All other physical quantities and formulas are derived from these basic quantities.